



Immobilization of Cu-MOF with SiO₂ aerogel in a marine coating to enhance its long-term anti-biofouling performance

Liuqin Zhang^{a,b}, Huali Li^c, Gege Zhang^a, Qiuping Li^d, Fa-Qian Liu^{a,*}

^a School of Chemical Engineering and Technology, Sun Yat-sen University, Zhuhai, 519082, China

^b School of Materials Science and Engineering, Sun Yat-sen University, Guangzhou, 510275, China

^c Institute of Chemistry of Henan Academy of Sciences, Zhengzhou, 450046, China

^d Aviation Key Laboratory of Science and Technology on Structural Corrosion Prevention and Control, China Special Vehicle Research Institute, Jingmen, 448035, China

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ABSTRACT

Marine biofouling can cause economic, environmental and safety issues. Here the Cu-MOF/SiO₂ aerogel@PDMS coatings were prepared to prevent the colonization of fouling organisms. The Cu-MOF-74 which can continuously provide copper ions was successfully synthesized through a simple method. SiO₂ aerogel was employed to immobilize Cu ions (Cu-MOF) by the formation of -Si-O-Cu-O-, thereby achieving the goal of more stable release of copper ions from the coating. And the improvement of coating hydrophobicity also helps to enhance the stability of MOFs. The Cu-MOF/SiO₂ aerogel@PDMS coating exhibits significant long-term antibacterial performance, with a maximum antibacterial rate of 99.5 %. And it can enhance the long-term static anti-biofouling performance of PDMS in the local sea by effectively preventing the colonization of macrofouling organisms (e.g. barnacles, calcareous tubeworms).

1. Introduction

Marine biofouling caused by those settlement of unwanted biological matter has aroused global concern due to its serious negative impact (e. g., financial losses, health risks) on marine industries [1–3]. Biofouling formation typically starts with the adsorption of organic molecules and particles onto surfaces within minutes, and later colonised by bacteria, diatoms and protozoa to form a biofilm [4,5]. Successfully settled bacterial communities may bring the successive adhesion of micro-algae (e. g., diatoms, cyanobacteria), fungi and protists, which promotes the colonization of larger fouling organisms or macrofouling [6]. In view of this, we can study the antifouling performance of the coating at any stage of the formation of biofouling.

Among various antifouling strategies, the antifouling coating has been proved to be the most effective approach. As is well known, Cu₂O is the most common and effective bactericide after the ban on organotin. However, there are still many defects of commercial Cu₂O-based antifouling coatings such as agglomeration, oxidation, short duration and high dosage [7]. Recently, metal organic frameworks (MOFs) are reported for their antibacterial potential. MOFs can be used as the source of metal ions which are stable through the formation of chemical bonds with organic ligands, and then can control the release of biocidal agents

[8–10]. In addition, compared with Cu₂O and those natural or synthetic materials loaded with metal ions, MOFs have an outstanding advantage, that is, the uniform distribution of metal active sites [9–11]. This enables MOFs to exert antibacterial effects not only by releasing metal ions, but also through their abundant active open metal sites (acting as the oxidizer to destroy the cell wall and plasma membrane) [12]. Therefore, MOFs have the potential to prepare antimicrobial agents with specific requirements. However, due to long-term immersion of marine antifouling coatings in seawater, it is possible that at a certain time period, MOFs materials cannot resist the attack of seawater, resulting in a large amount of collapse of their structure, which will cause the release of a large amount of antifouling agents [13]. It is expected that improving the hydrophobicity of the coating will help enhance the stability of MOFs.

Polydimethylsiloxane (PDMS) is a typical material used for preparing hydrophobicity and low surface energy silicone antifouling coatings, which is difficult for fouling organisms to firmly adhere to the surface due to the smooth surface, low surface energy (23 mJ/m²), low modulus of elasticity (2 MPa), water repellency, high structural mobility and controlled release [1,7,14]. However, the use of low surface energy antifouling coatings alone during static operation of ships or port berthing inevitably results in the attachment of some fouling organisms,

* Corresponding author.

E-mail address: liufq7@mail.sysu.edu.cn (F.-Q. Liu).

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leading to scaling and colonization [15]. These undesirable consequences have stimulated the development of PDMS functionalization strategies [16,17]. Filling biocide (e.g., MOFs, agents from sessile marine organisms, microorganisms) is one of the methods. This strategy can not only improve the antibacterial performance of low surface energy coatings, but also enhance the water repellency of the aforementioned MOFs. According to our investigation, MOFs/PDMS hybrid materials have extensive research in fields such as tissue engineering [18,19], separation [20,21], gas storage [22,23], water purification [24], biosensor [25] and catalysis [26]. However, there is still relatively little research and application of this composite material in marine anti-fouling coatings.

Herein, we developed a marine antifouling coating consisting of a typical Cu-MOF (Cu-MOF-74), SiO₂ aerogel and PDMS. The filling of Cu-MOF gives the coating continuous antifouling performance as it could continuously provide copper ions (Cu ions). SiO₂ aerogel is employed to immobilize Cu ions (Cu-MOF) by the formation of -Si-O-Cu-O-, thereby alleviating the massive release of copper ions caused by a large number of structural collapse of MOFs due to long-term immersion in seawater. The crystallinity, topology, size of Cu-MOF, and Cu site accessibility of Cu-MOF/SiO₂ aerogel hybrid material are retained after PDMS coating. Remarkably, the introduction of PDMS improves the hydrophobicity of the coating, which can protect the MOFs in the coating from severe attacks by seawater and significant degradation. As a result, the PDMS coated Cu-MOF/SiO₂ aerogel (Cu-MOF/SiO₂ aerogel@PDMS coating) exhibits significantly improved stable copper ion release performance and enhanced long-term static anti-biofouling efficiency of PDMS.

2. Materials and methods

Materials and Reagents. 2,5-dihydroxyterephthalic acid (DHTP, 98 %), tetraethyl orthosilicate (TEOS, 99 %), ethanol (EtOH, AR), methanol (MeOH, AR) and glutaraldehyde (AR) were purchased from Aladdin Reagent Co., Ltd. Copper(II) acetate (Cu(OAc)₂, 98 %), poly (acrylic acid) (viscous liquid, 30 %, average molecular weight: ~3000) were provided by Macklin Reagent Co., Ltd. Acetone (AR) and nitric acid (HNO₃, GR) were provided by Guangzhou Chemical Reagent Factory. The oligomer polydimethylsiloxane (PDMS, Sylgard 184) and curing reagent were obtained from Dow Corning. Artificial seawater (ASW) was fabricated as GB/T 7790-1996. Copper standard solution (1000 ppm) was obtained from Guangzhou Testing Technology Co., Ltd. Deionized water (DI water) and *Escherichia coli* (*E. coli*, DH5 α , DingGuo) were provided by local laboratory. No additional processing was done for the materials. The slides (Sail Brand) used as coating substrate were respectively sonicated in acetone and aqueous solution for 20 min and then transferred to absolute ethanol before use.

Preparation of Cu-MOF. Cu-MOF-74 has been proven to have antibacterial properties in membrane materials, therefore it is chosen as the Cu-MOF material to improve the antibacterial and antifouling performance of the fouling release marine coating. The Cu-MOF-74 was synthesized by a facile method. Firstly, 2 mmol of Cu(OAc)₂ was dissolved in 10 mL of MeOH. Then another 5 mL of MeOH was used to dissolve 1 mmol DHTP. After that, the DHTP solution was added dropwise into the Cu(OAc)₂ solution. The mixture was magnetically stirred for 24 h at ambient temperature and then reddish-brown crystalline solid containing Cu-MOF-74 was obtained. After centrifugation, the solid was immersed in 20 mL fresh MeOH for 24 h and 4 times to fully wash away impurities. At last, the as-prepared Cu-MOF-74 was collected by centrifugation and dried in an 85 °C vacuum for 5 h.

Preparation of SiO₂ aerogel. 2 mL of polyacrylic acid was added into the water (2.16 mL) and ethanol (5.84 mL) mixed solution. After mixed well, 4.4 mL of TEOS was added and the mixed solution was stirred magnetically for 1 h. After that, it was transferred into a 20 mL teflon-sealed autoclave and heated at 60 °C for 24 h. Then, the silica hydrogel was obtained and transferred into a 50 mL beaker. After soaked in 20 mL ethanol for 3 days, it was repeat washed with ethanol to remove

unreacted impurities. Finally, the obtained silica hydrogel was put into an oven at 60 °C for aging and drying to obtain the SiO₂ aerogel (BET pore size: 1.76 nm).

Preparation of Cu-MOF/SiO₂ aerogel hybrid material. To obtain the Cu-MOF/SiO₂ aerogel hybrid material (Cu-MOF-74/SiO₂ aerogel hybrid material), 0.5 g of SiO₂ aerogel and 0.4 g of Cu(OAc)₂ were dissolved in 10 mL of MeOH. And the mixture was magnetically stirred for 4 h to fully mix the copper ions and SiO₂ aerogel. Then 0.2 g of DHTP was dissolved in another 5 mL of MeOH and then added dropwise into the above solution. The subsequent processing was the same as the preparation of Cu-MOF.

Preparation of Cu-MOF/SiO₂ aerogel@PDMS coating. Cu-MOF/SiO₂ aerogel hybrid material (3 wt% or 6 wt% with respect to the weight of the monomers) was dispersed into PDMS (10 wt% of curing reagent). After stirring evenly, the mixture was placed in a refrigerator at 4 °C for 30 min to eliminate bubbles. And then the materials were coated on glass slides (2.5 cm $\times$ 2.6 cm), which were put into a 60 °C oven to accelerate solidification and obtain marine antifouling. The coatings are marked as Blank (only PDMS), Control (3 % Cu-MOF-74 in PDMS), Cu + Si 3 % coating (1.5 % Cu-MOF-74 and 1.5 % SiO₂ aerogel in PDMS) and Cu + Si 6 % coating (3 % Cu-MOF-74 and 3 % SiO₂ aerogel in PDMS), CuSi 3 % coating and CuSi 6 % coating (3 % and 6 % Cu-MOF/SiO₂ aerogel hybrid material in PDMS), respectively (See Table 1 for details).

Structural Characterization. The morphology of MOF-74, SiO₂ aerogel and the distribution of Cu on the coating cross-section were confirmed by scanning electron microscopy (SEM, regulus 8230). Chemical composition, and structure of Cu-MOF materials and coatings before or after immersion were confirmed by powder X-ray diffraction (XRD, Empyrean), X-ray photoelectron spectrometry (XPS, Thermo Fisher Nexsa) and fourier transform infrared spectroscopy (FT-IR, Nicolet 6700). The optical contact angle meter (OCA15EC, DataPhysics Instruments GmbH) was applied to assess the hydrophobicity/hydrophilicity of the coatings. The exudation rate of copper ions in the coatings was tested by atomic absorption spectroscopy (Thermo, SOLAAR ICE 3500).

Elastic deformation test. The CuSi 6 % coating (before and after immersion in local sea for 3.5 months) was used as a representative coating for nanoindentation experiments by a nanoindentation instrument (Hysitron TI 950). Here, a Berkovich indenter was used, with a indentation load of 15 mN and a loading speed of 1 mN/s, maintaining a maximum load of 5 s. We can obtain the elastic deformation resistance of the coating, as well as the stiffness, hardness, and other data of the coating through nanoindentation experiments (Table 2).

Copper ions Leaching Trial. The coatings was immersed in ASW in a big crystal dish and stirred for 1 day. And then every coating was immersed in 50 mL of ASW under magnetic stirring for 12 h, respectively. After that, the leaching solution was obtained and then diluted (psalt $\leq$ 500 mg/L) by 0.5 % nitric acid to detect and record the metal ions in the coating for 1 day exudation concentration. Afterwards we replace the remaining solution in the big crystal dish (replace ASW every 7 days) and use the same method to detect the concentration of metal

Table 1

The contents of Cu-MOF-74 in different coatings.

Samples	W (PDMS)/g	W(Cu-MOF-74)/g	W (SiO ₂ aerogel)/g	W(Cu-MOF-74/SiO ₂ aerogel hybrid material)/g
Blank	2.5	0	0	0
Control	2.5	0.077	0	0
Cu + Si 3 % coating	2.5	0.039	0.039	0
Cu + Si 6 % coating	2.5	0.077	0.077	0
CuSi 3 % coating	2.5	0	0	0.077
CuSi 6 % coating	2.5	0	0	0.155

Table 2

Mechanical properties of the CuSi 6 % coating before and after immersion in local sea for 3.5 months.

Elastic Modulus (GPa)		Hardness (GPa)		Stiffness ($\mu\text{N}/\text{nm}$)	
before	after	before	after	before	after
0.0052	0.0062	0.0013	0.0019	0.62	0.61

ions in the solution after the coating was immersed for 3, 7, 10, 14 and 21 days, respectively.

Hydroxyl Radicals ($\cdot\text{OH}$) Assays. Electron paramagnetic resonance instrument (EPR, Bruker A300-10/12) was used to test the generation of $\cdot\text{OH}$ as a representative of reactive oxygen species (ROS) under light and dark conditions.

Antibacterial Assays. The precultured *E. coli* (1 CFU) was

transferred into 10 mL of Luria-Bertani (LB) medium, cultured for 8 h at 37 °C, 180 rpm. After that, the solution was centrifuged to leave the bacteria which was resuspended in 10 mL of 0.9 % NaCl solution. Next, 200 μL of the bacterial suspension was diluted 100 times with 0.9 % NaCl in a 50 mL sterile centrifuge tube, and then coating (after 1, 3, 7, 10, 14, 21 days immersing in ASW, respectively) was added. The centrifuge tubes were put in the same shaker 5 h (day1-day14) or 12 h (day21) (37 °C, 180 rpm). After that, the coatings were taken out. Then, 800 μL of bacterial solution was transferred from different centrifuge tubes and diluted in different multiples ($10\times$, $100\times$ and $1000\times$). The number of surviving colonies were counted by the plate counting method (100 μL) and recorded as the number of surviving bacterial colonies after 24 h. Repeat the above process three times.

Morphology of *E. coli*. The precultured *E. coli* (1 CFU) was transferred into 10 mL of LB medium, cultured for 24 h at 37 °C, 180 rpm. The prepared *E. coli* were resuspended with 0.9 % NaCl after centrifugation

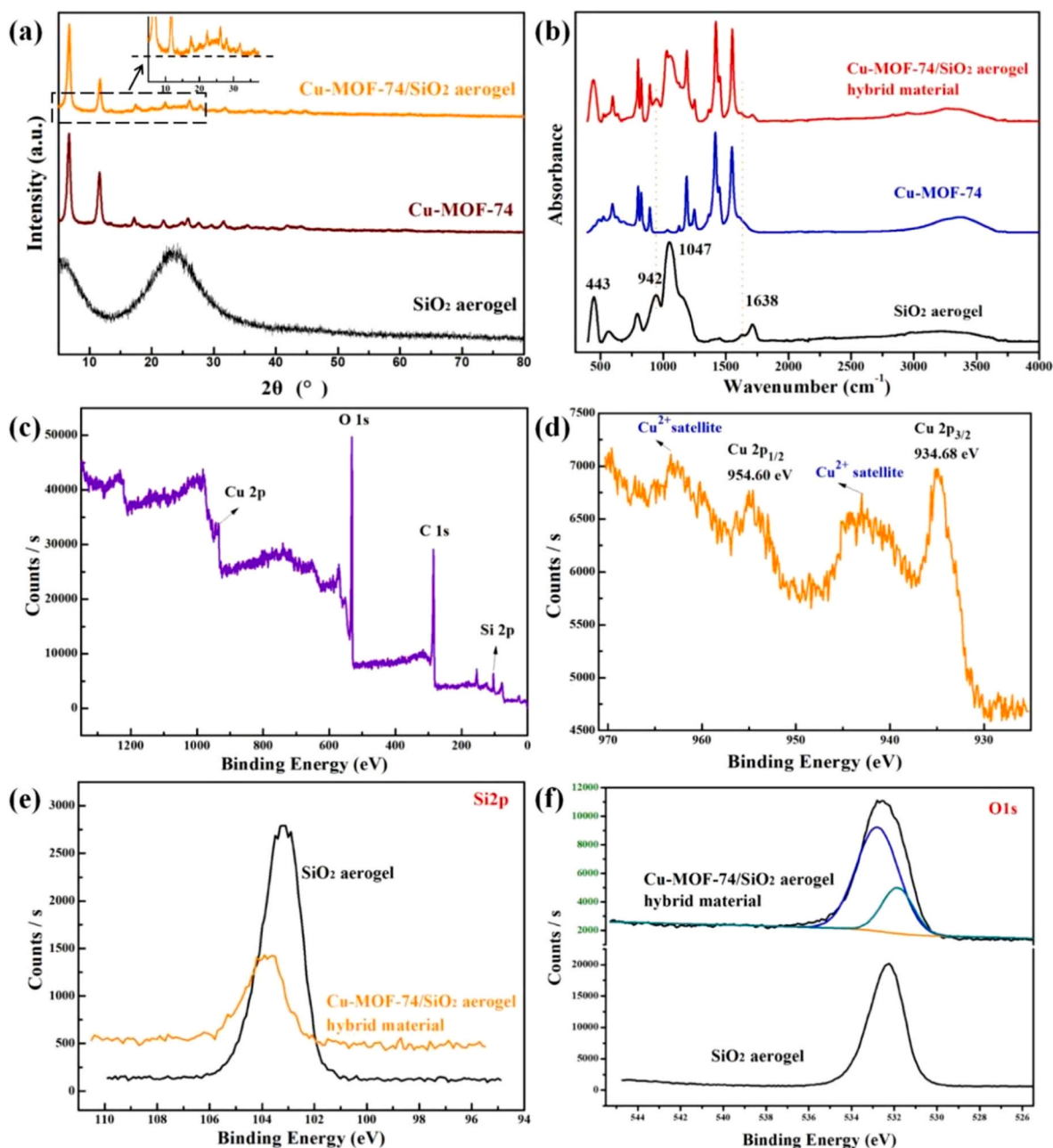


Fig. 1. (a) The XRD patterns and (b) FT-IR curves of Cu-MOF-74, SiO₂ aerogel and Cu-MOF-74/SiO₂ aerogel hybrid material. (c) XPS survey and (d) Cu2p spectra of Cu-MOF-74/SiO₂ aerogel hybrid material. (e) Si2p and (f) O1s of SiO₂ aerogel and Cu-MOF-74/SiO₂ aerogel hybrid material.

and then divided into four equal parts and put into the 50 mL sterile centrifuge tube respectively. Then the coatings were put into these sterile centrifuge tubes in which 0.9 % NaCl solution was added until the coatings were submerged. After shaking 24 h at 37 °C, 180 rpm, bacteria were separated and washed with 0.9 % NaCl solution and fixed with 3 mL of 2.5 % glutaraldehyde at 4 °C for 24 h. After that, the samples were washed with 0.9 % NaCl and dispersed in ethanol of different concentrations (50 %, 75 % and 100 %) for dehydration (15 min). Finally, their morphology was observed by SEM.

Locale Experiment. The coatings were immersed in local sea (Zhuhai, China) 1.5–2 m deep for 3.5 months to evaluate the ability of coatings to prevent macrofouling organisms' colonization under static conditions (compared to high-speed navigation of ships). Specifically, after retrieving the samples from the sea, the sludge on them was slightly cleaned with tap water to facilitate observation of biofouling. It should be noted that during sample processing, the opening of the faucet will not be changed after adjusting the faucet outlet, and the distance between each sample and the faucet outlet should be consistent to ensure that each coating was washed by the same pressure of water flow.

3. Results and discussion

3.1. Characterization of Cu-MOF/SiO₂ aerogel hybrid material

From XRD patterns (Fig. 1a), two typical peaks at $2\theta = 6.7^\circ$ and 11.8° are observed in Cu-MOF-74, which is consistent with the diffraction peaks of the reported Cu-MOF-74 [27–31]. This indicates that the Cu-MOF-74 was successfully synthesized. The XRD pattern of the SiO₂ aerogel shows a broad diffraction peak at 20° – 30° , which is associated with the spacing of Si atoms and angle of the group Si–O–Si [32]. The diffuse peak suggests that the aerogel has low crystallinity, which is a amorphous silica material. Moreover, it is observed there are two characteristic diffraction peaks at $2\theta = 6.7^\circ$ and 11.8° in the XRD spectra of Cu-MOF-74/SiO₂ aerogel hybrid material. Meanwhile, it shows a dispersing diffraction peak at $2\theta = 20^\circ$ – 30° in the magnified spectrum of Cu-MOF-74/SiO₂ aerogel hybrid material (the embedded graph in Fig. 1a–S1), which corresponds to the dispersion peak of SiO₂ aerogel at that location. And the peak baseline at $2\theta \approx 5^\circ$ is not as flat as the spectrum of pure Cu-MOF-74, also attributed to the presence of SiO₂ aerogel. These indicate that Cu-MOF-74 was synthesized in the presence of SiO₂ aerogel without altering its topological structure.

FT-IR spectra (Fig. 1b) were recorded for further studying the incorporation of Cu-MOF-74 and SiO₂ aerogel. The three strong and wide absorption peaks at 1047 cm^{-1} , 942 cm^{-1} and 443 cm^{-1} are the stretching and bending vibration absorption peaks of Si–O–Si bonds, and the absorption peak at 1638 cm^{-1} is assigned to Si–OH vibrations. These conform the form of SiO₂ aerogel through the connection of Si–O bonds and the presence of SiO₂ inorganic phase in hybrid materials. Moreover, the XPS survey spectrum shows that there are Si and O atoms, and the Si2p at binding energy is $> 103\text{ eV}$ in Si2p XPS spectrum, indicating that the Si in the sample is present as SiO₂ (Fig. 1e–S2). For the FT-IR curve of Cu-MOF-74, it has a good agreement to the reported MOF-74 between band position and relative intensity [30,33]. Compared with pure SiO₂ aerogel, the peaks near 942 cm^{-1} and 1638 cm^{-1} are greatly weakened, and the peak at 1042 cm^{-1} splits into two absorption peaks in the hybrid material, which is related to the reaction of Si–OR/Si–OH groups in SiO₂ aerogel with Cu ions to generate Si–O–Cu groups. In addition, it can be seen from the XPS survey spectrum of Cu-MOF-74/SiO₂ aerogel hybrid material there are C, O, Si, Cu atoms (Fig. 1c), and the existence of the satellite peaks in the Cu2p XPS spectrum (Fig. 1d) suggests that copper in the hybrid material mainly exists in divalent form (Cu²⁺). Furthermore, the counts values (intensities) of the Si2p and O1s peaks of the Cu-MOF-74/SiO₂ aerogel hybrid material are smaller than that of the pure SiO₂ aerogel (Fig. 1e and f). And the O1s spectrum of hybrid material shows a peak at the binding energy of 532.2 eV – 532.9 eV , which is assigned to the Si–O–Si. More importantly, a shoulder peak appears at

the binding energy of 531.5 eV – 531.9 eV , indicating that copper ions have entered the Si–O framework (Fig. 1f). The XPS results further prove the binding between Cu and Si–O– to form Si–O–Cu [34,35].

It is known that the observed particles in Cu-MOF-74 have sizes far below the micrometer level and are composed of agglomerates/aggregates formed by a large number of nanocrystals (Fig. 2a) [28,30]. When observing the morphology of the Cu-MOF-74/SiO₂ aerogel hybrid material, it can be seen from Figs. 2c–1 that the Cu-MOF-74 is distributed on SiO₂ aerogel particles. And it is found that some of the particles are the same as pure Cu-MOF-74 particles (Figs. 2c–4,d), another are the large particles consisting of SiO₂ aerogel block and Cu-MOF-74 (Fig. 2c–, c-3,d). The former is that smaller SiO₂ aerogel particles or free silicon ions first combine with copper ions, and then react with DHTP to form MOF crystals with the same shape as pure Cu-MOF-74 particles. However, due to the presence of heterogeneous central ions, the MOF aggregates formed are relatively loose and not as tight as pure Cu-MOF-74, which is consistent with our previous research [13]. The latter is that Cu-MOF-74 particles embedded in SiO₂ aerogel are generated by the reaction of copper ions in pores or on the surface of those large SiO₂ aerogel particles with DHTP. The large particles of SiO₂ aerogel have rough surfaces and pits (Figs. 2b–1,b-2), which are conducive to the adsorption of copper ions and the subsequent insertion of Cu-MOF-74. The Mapping images of Cu-MOF-74/SiO₂ aerogel hybrid material further confirm the above inference. Among them, the large particle at position 2 (P2) exhibits a clear distribution of Cu and Si due to the smaller Cu-MOF-74 grown on it. On the large particle at position 3 (P3), Cu-MOF-74 with larger particle size grows, and then the SiO₂ aerogel underneath is masked, thus only showing a small amount of Si distribution. However, the contour of the particle at P3 can still be seen in the Si Mapping image. The above indicate that Cu-MOF-74 and SiO₂ aerogel in the Cu-MOF-74/SiO₂ aerogel hybrid material have formed a good combination.

3.2. Characterization of coatings

In the XRD patterns, it can be seen that all coatings before immersion in local sea for 3.5 months, including the Blank (only PDMS) coating, have a peak near $2\theta = 11.8^\circ$ (Fig. 3a). However, there is a weak peak near $2\theta = 6.8^\circ$ in the Control (3 % Cu-MOF-74 in PDMS) coating (Fig. 3b), which is more pronounced in the Cu + Si 6 % coating (Fig. 3c). For the CuSi 6 % coating, the characteristic peak at $\sim 6.8^\circ$ is observed (Fig. 3d), but it is extremely weak. However, the peak at $\sim 11.8^\circ$ is higher and sharper than that in the Blank coating. After 3.5 months immersion, it is observed that the peaks at 5° – 6.8° and 11.5° – 11.8° are more dispersed in the Blank coating, while the peaks at the corresponding positions of the coatings containing Cu-MOF materials are more pronounced and sharper. This may be because as the immersion time increases, the Cu-MOF materials in PDMS gradually migrate to the surface layer (Fig. 5c and d), allowing for better acquisition of Cu-MOF-74 information during XRD detection. These results indicate that Cu-MOF materials have been successfully incorporated into PDMS without changing the topological structure, but their contents on the surface of PDMS are relatively low.

In addition, it was measured that the water contact angle of the Blank coating is 117° . After adding Cu-MOF materials, the contact angle of the coatings decreases by $1\sim 2^\circ$ (Fig. 4). This is because Cu-MOF-74 contains abundant –COOH and –OH, which can be inferred from the surface of CuSi 6 % coating containing 0.35 % Cu (Fig. 5a). Nevertheless, it also indicates that the addition of Cu-MOF materials does not have a significant impact on the hydrophobicity of the coatings. Moreover, the water contact angles of the coatings were measured after Locale Experiment. The result shows that the water contact angles of the coatings remain above 113° after soaking in natural seawater for 3.5 months, indicating that the coatings can maintain hydrophobicity for a long time in harsh marine environments. The reason for the decrease in contact angle of the coating after immersion is that there are some

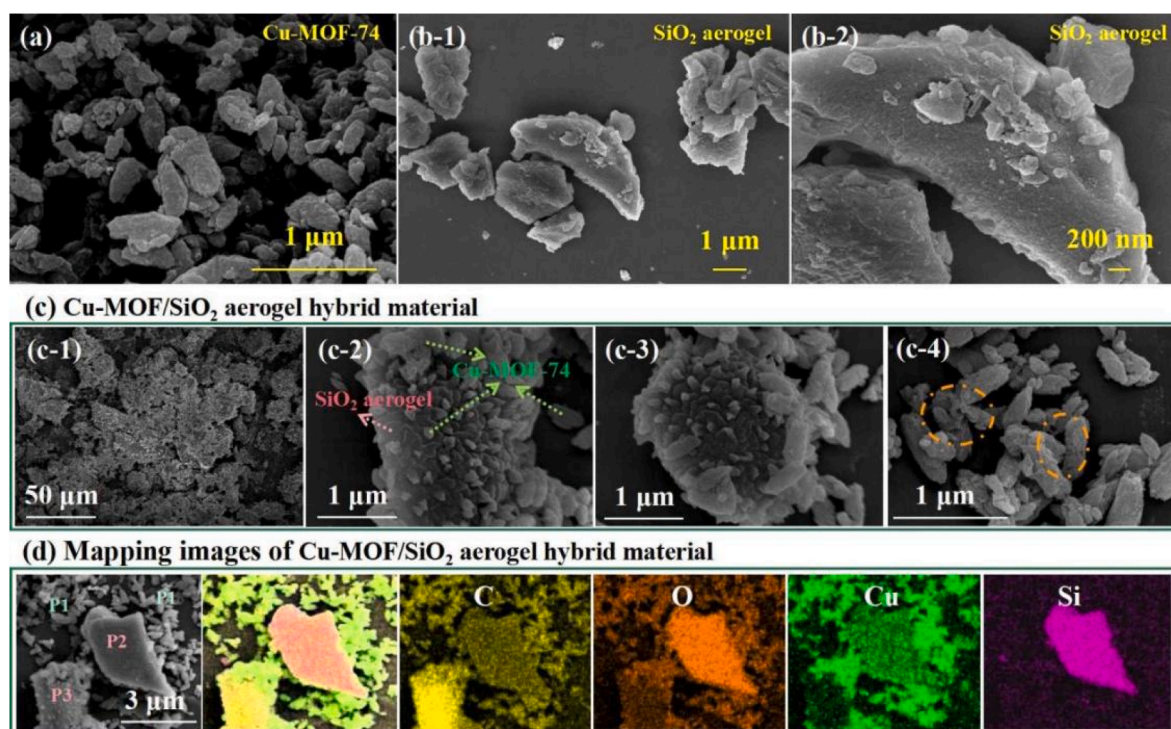


Fig. 2. The SEM images of (a) Cu-MOF-74, (b) SiO₂ aerogel and (c) Cu-MOF-74/SiO₂ aerogel hybrid material. (d) The Mapping images of Cu-MOF-74/SiO₂ aerogel hybrid material.

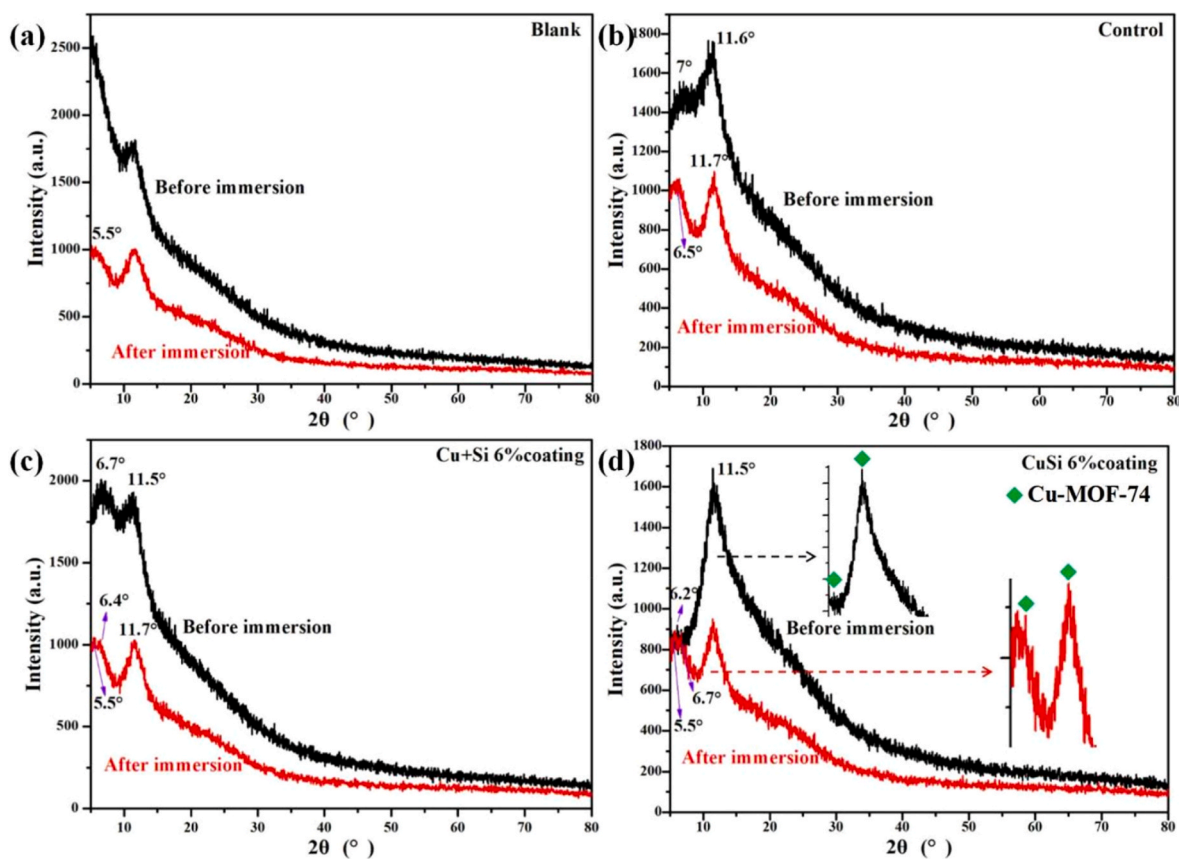


Fig. 3. The XRD patterns of coatings before and after immersion in local sea for 3.5 months (the green diamonds represent the characteristic peaks of Cu-MOF-74).

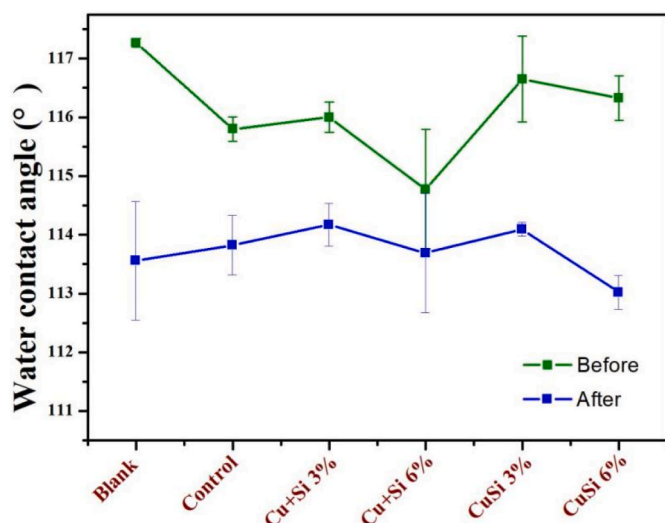


Fig. 4. Water contact angles of the coatings before and after immersion in local sea.

marine sediments on the surface of the coating (after removing obvious marine organisms) (Fig. 5b). In addition to the main elements of C, O, and Si, the coating surface also contains a small amount of K, Ca, Mg and

other elements, which are inferred to be from the sea salts and sludge. These sediments affect the detection of coating contact angle and Cu content. And the mapping images of the coating cross-section show that a large amount of Cu element is deposited at the bottom of PDMS before immersion, and then distributed more evenly after immersion (Fig. 5c and d), which corresponds to the results of XRD. However, after 7 days immersion in ASW, the XRD spectrum of Cu-MOF-74/SiO₂ aerogel hybrid material without PDMS treatment shows that it transformed into other substances, and the color changed from reddish brown to dark green (Fig. S4a and c). On the contrary, the hybrid material with PDMS treatment maintains a reddish brown color and has two XRD characteristic peaks at $\sim 6.7^\circ$ and $\sim 11.8^\circ$ even immersion in local sea for 3.5 months (Figs. 3d and 8, S4b). Therefore, the hydrophobicity can help improve the hydrolytic stability of MOFs and then improve their protection efficiency and life span [36].

Furthermore, we conducted 6 cyclic nanoindentation experiments with 15 mN at the same location of the CuSi 6 % coating. Figs. 6a–1 shows that after applying pressure to the coating (before immersion), the depth of coating indentation increases with larger pressure, but after pressure relief, the coating gradually recovers (more than 88 % of deformation can be restored) and Figs. 6a–2 displays that the time required for coating recovery is very short. The test results show that the coating has good elastic deformation recovery ability after multiple pressurization and depressurization before immersion in sea water. Moreover, after the nanoindentation experiment/uninstallation, no indenter shape is found on the coating surface (Fig. S5). Besides, the

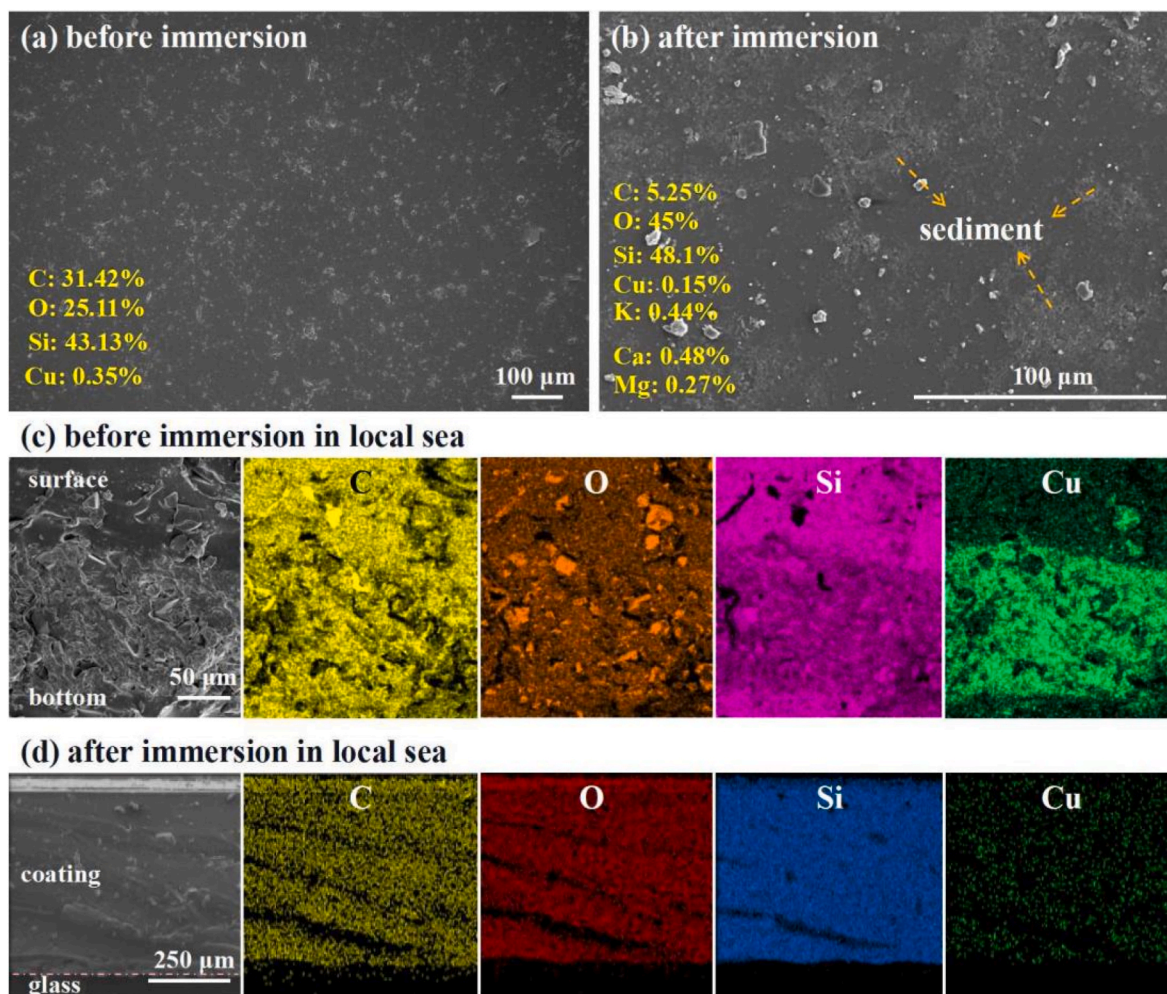


Fig. 5. The surface morphology of CuSi 6 % coating (a) before and (b) after immersion in local sea for 3.5 months. The cross-section morphology and mapping images of CuSi 6 % coating (c) before and (d) after immersion in local sea.

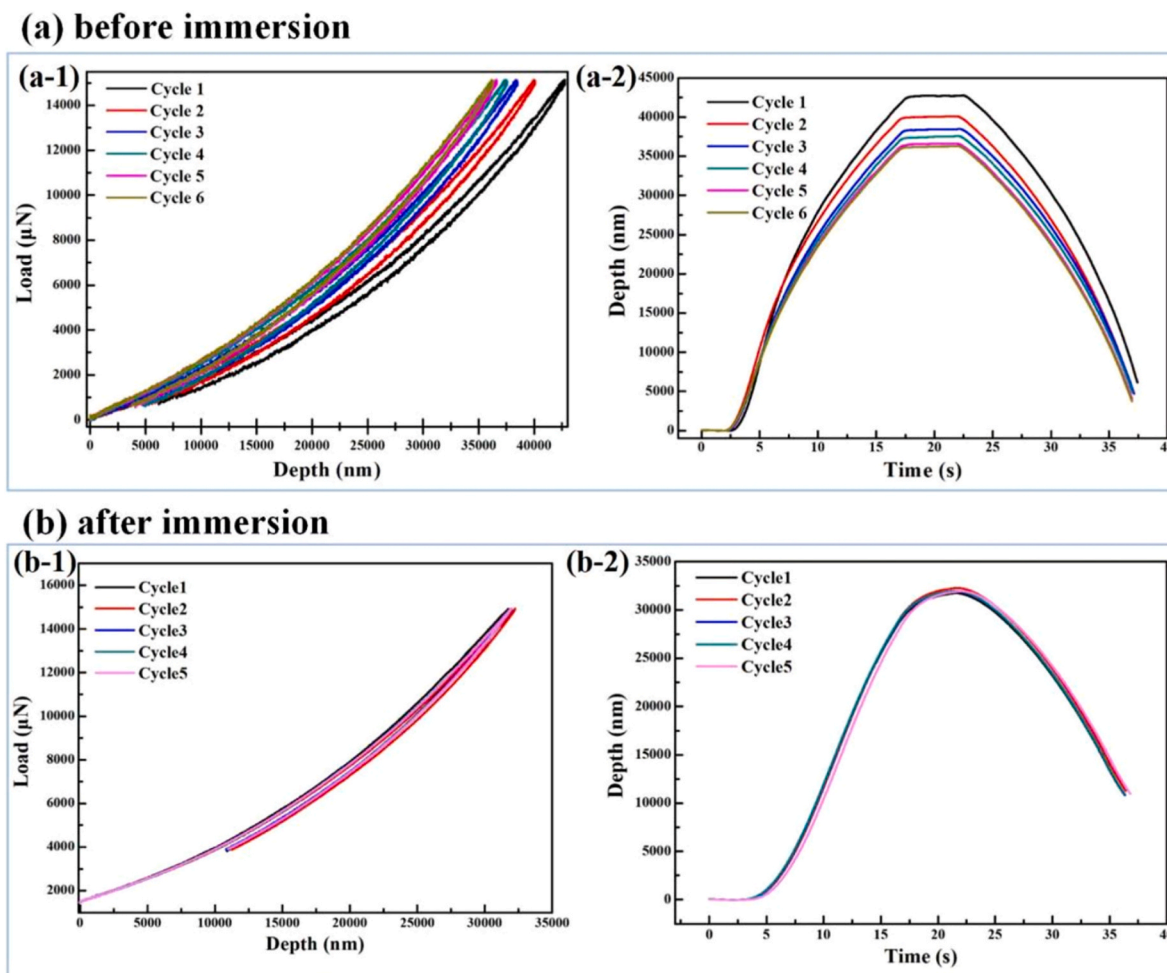


Fig. 6. Nanoindentation experiments including Load-Depth and Depth-Time curves of the CuSi 6 % coating (a) before and (b) after immersion in local sea for 3.5 months.

elastic modulus (0.0052 GPa) and stiffness (0.62 $\mu\text{N}/\text{nm}$) of the CuSi 6 % coating are quite small (Table 2) (of the PBMA we previously prepared was 343.88 GPa, and the stiffness was 56555 $\mu\text{N}/\text{nm}$ [13]). The smaller the elastic modulus and stiffness, the easier it is to recover from deformation. These further indicate that the coating has excellent elastic recovery ability. That is to say, the prepared coating has superior resistance to the impact of sediment in the ocean, which can be further confirmed by the results of later nanoindentation experiments on the coating after immersion in local sea for 3.5 months. From Fig. 6b, although the elastic deformation recovery time of the coating has been extended, its elastic modulus (0.0062 GPa), stiffness (0.61 $\mu\text{N}/\text{nm}$) and hardness (0.0019 GPa) have not changed significantly (Table 2). And Fig. 8 shows that the coating has not been damaged. In short, the as-prepared coating is able to maintain good mechanical properties after immersion in sea water for a long time.

3.3. Anti-biofouling property

3.3.1. Evaluation of long-term antibacterial property

Antibacterial experiment lasting three weeks was conducted to assess the long-term antibacterial property of the coating. As can be seen from Fig. 7, the fabricated coatings show remarkably good antibacterial properties. The first week of antibacterial experiments display that the coating has excellent antibacterial effects. The antibacterial rates almost always above 90 %, with a maximum of 99.5 % (day1, day3 and day7). In the second week of the experiment, it is found that the antibacterial activity of the coatings decrease, and the antibacterial rates of the

coatings containing Cu-MOF-74 are 70 %–80 % (day10) and 33 %–58 % (day14). Considering that the coatings were immersed in artificial seawater (ASW) for a long time, the salts in the ASW will precipitate and cover the surface of the coatings, making it difficult for copper ions in the coating to seep out, and also masking the fixed Cu-MOF on the surface of the coatings to expose Cu sites (day10 and day14). In addition, the precipitation increases the roughness of the coating surface making it easier for bacterial cells and organic molecules to adhere; and the coatings containing MOF can cause bacterial cells to rupture and leak their contents (polysaccharides, proteins, etc), providing nutrients for newly added bacteria [37]. Therefore, on the 14th day, the antibacterial effect of PDMS is slightly better than that of MOF coatings. In view of this, in the experiment on the 21st day (day21), we extended the interaction time between the coatings and the bacterial solution from 5 h to 12 h, allowing the salt settling on the coating surface to dissolve as much as possible, promoting the smooth release of copper ions from the coating and exposing the Cu sites on the coating surface. After that, the antibacterial rates of the coatings have indeed increased, reaching over 60 %, with a maximum of 83 %. Overall, the prepared coating has good long-term antibacterial property.

3.3.2. Anti-macrofouling organisms settling performance

As shown in Fig. 8b, there are many barnacles and calcareous tubeworms growing on the glass slide, while no barnacles are found on other coatings and the size of the growing calcareous tubeworms is much smaller than that on the glass slide. This indicates that fouling organisms are not as fond of colonizing on organic silicon coatings.

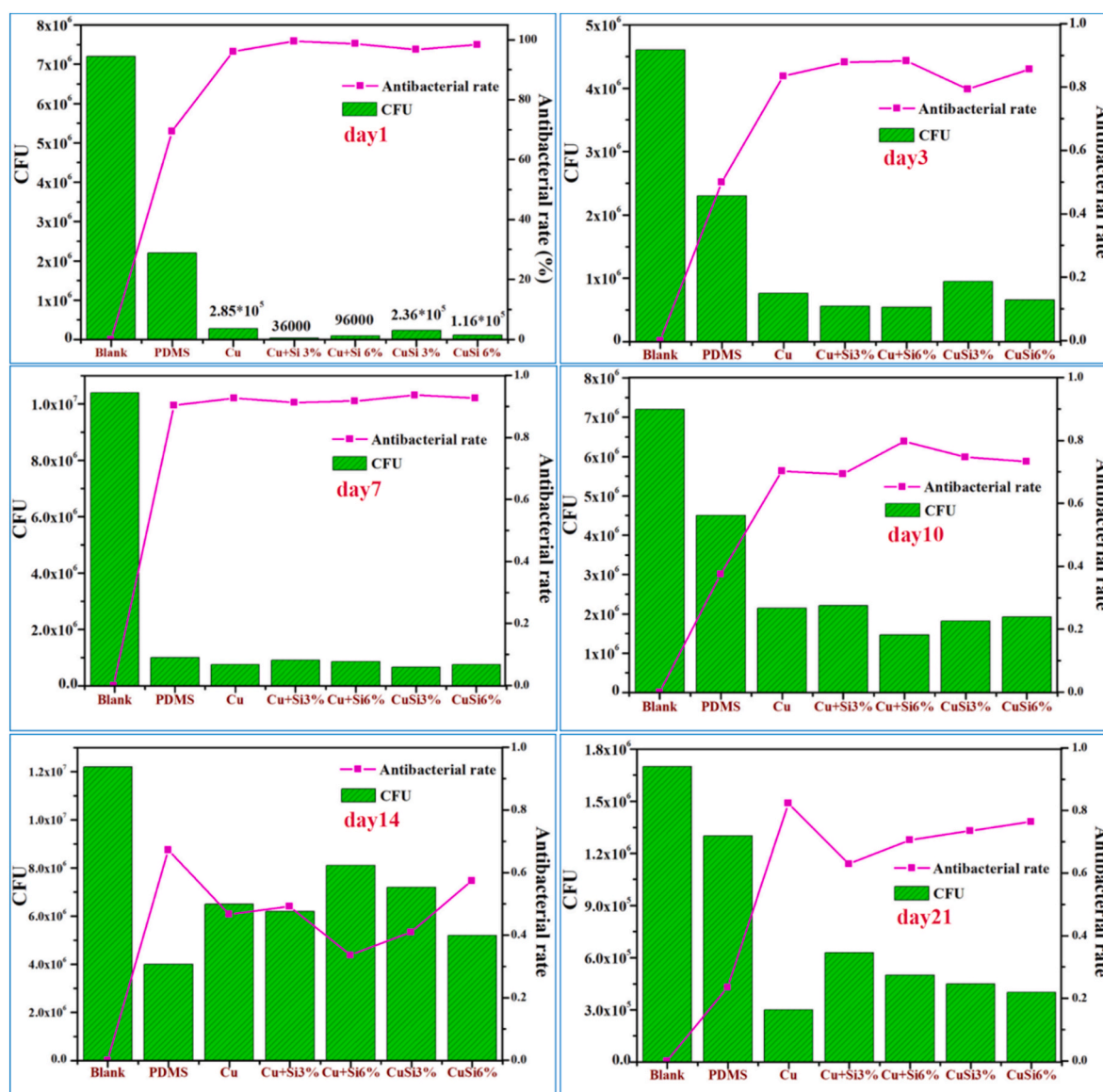


Fig. 7. Numbers (CFU: colony forming units) and antibacterial rate of surviving bacteria (*E. coli*) clones on culture plates after contacting with different coatings.

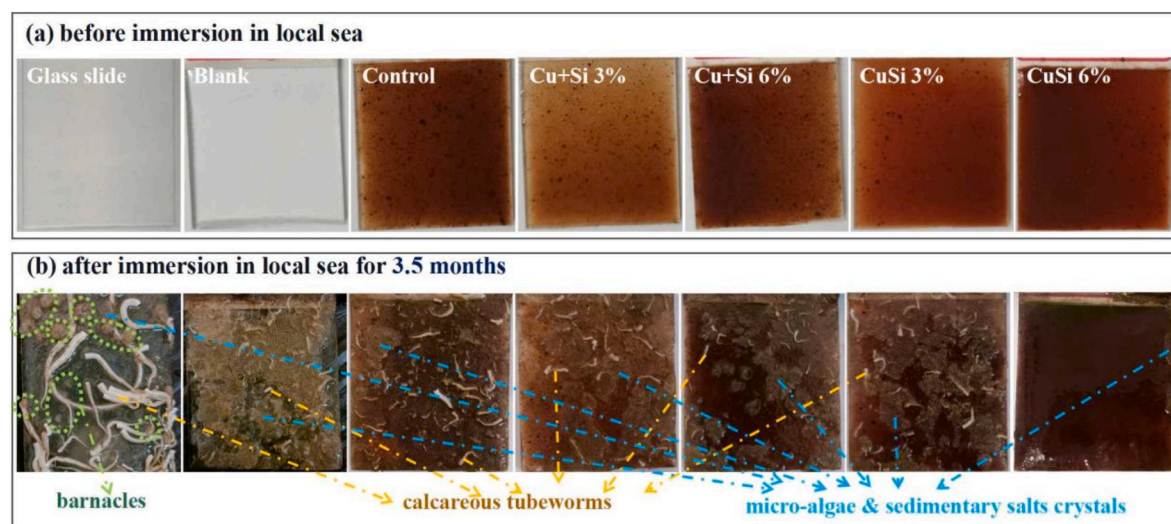


Fig. 8. Colonization of macrofouling organisms on coatings (The coatings in Fig. 8a and b correspond to each other).

Furthermore, compared to the following four coatings (Cu + Si 3 %, Cu + Si 6 %, CuSi 3 % and CuSi 6 %), the Blank and Control coatings exhibit a denser growth of calcareous tubeworms, which confirms that the Cu-MOF-74 and SiO₂ aerogel composite materials can help enhance the anti-biofouling performance of the coatings. Moreover, there are fewer tubeworms on the Cu + Si 6 % coating compared to Cu + Si 3 %, and there is almost no tubeworm attachment on CuSi 6 % coating, which indicates that the Cu-MOF-74 content (introduce more open Cu sites) has an impact on the anti-biofouling performance of the coating and the as-prepared Cu-MOF-74/SiO₂ aerogel hybrid material has better anti-biofouling effects than simple mixed materials. However, due to biodiversity (different surface properties of biological cells) and substrates surface properties, organisms tend to selectively attach. For example, Krasowska et al. reported that hydrophobic microorganisms favor hydrophobic material surfaces while those hydrophilic microorganisms favor hydrophilic material surfaces [38,39]. Domínguez et al. also reported that the cells with less polar components in their wall (*Synechococcus* sp.) are easier to adhere to the lower polar component surface with great adhesion force [40]. Therefore, a coating may not completely eliminate the attachment of all organisms, which is probably the reason why a small amount of fouling organisms adhered to the coatings we prepared (although they have a certain inhibitory effect). In addition, the deposition of network-like substance is observed on the coatings, which may be a mixture of micro-algae (specific analysis in section 3.4.3), sea salts and sludge. This confirms our speculation in the antibacterial research mentioned above about the deposition of ASW salts on the coatings. Above all, the addition of Cu-MOF-74/SiO₂ aerogel hybrid material improves the static anti-biofouling effect of organic silicon coatings.

Antimicrobial agents are constantly developing with time going by. The antimicrobial agents reported can be classified into inorganic (e.g., Ag, Zn, Cu, Fe and their oxides and nanoparticles, carbon nanotubes, graphene) [13,41–44], organic antimicrobial agents (e.g., quats, calcium formate, phthalocyanine compound, ConShield) [42,45,46] and biological extract (venoms of animals such as snakes, scorpions, bees, and spiders; terpenes, polyphenols, and organic acids from plants; bacteriocins and antibiotics) [47]. The above materials each have their own advantages and disadvantages, such as the high temperature resistance but toxicity of inorganic antimicrobial agents, broad spectrum of killing activity but unstable of organic antimicrobial agents, and environmentally friendly but low yield and high cost of biological extract [42,48]. Among the antimicrobial agents, MOFs materials have the following advantages: (1) MOFs can be used as a reservoir of metal ions, which are atomically dispersed and increasing atomic utilization [8,49]; (2) the release of metal ions in MOFs is slow and toxic components degraded is smaller than from the pure component [8,50]; (3) MOFs with high specific surface area and porosity can act as carriers for other antimicrobial agents or drugs [51,52], which endows MOFs materials with multiple antibacterial mechanisms; (4) they can be prepared into three-dimensional (3-D), two-dimensional (2-D), and one-dimensional (1-D) materials as needed, improving antibacterial performance while enhancing the mechanical properties, impermeability, and other properties of coatings, concrete, etc. [53–55].

3.4. Anti-biofouling mechanisms of coatings

3.4.1. The release rate of copper ions

The release rate of copper ions will affect the anti-fouling performance and service life of the coating. Here, we continuously tested the release concentrations of various coatings in simulated marine environment for 21 days. As shown in Fig. 9, Cu-MOF (Control coating) can indeed continuously release copper ions, but its release rate is not stable enough, fluctuating between 0.03–0.14 mg/L. After adding a small amount of SiO₂ aerogel (Cu + Si 3 % coating), the release of copper ions in the coating decreases first and then increases. When the content of SiO₂ aerogel increases (Cu + Si 6 % coating), the stability of copper ions

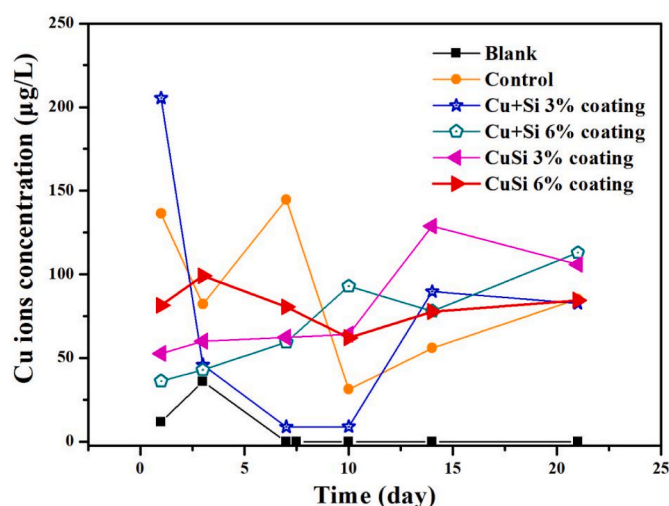


Fig. 9. Concentration of copper ions released from different coatings after immersed in the ASW for different days.

released by the coating increases. This is because the bonds of -Si-O- increases with the increase of SiO₂ aerogel, so that more copper ions can bond with -Si-O- to synthesize -Si-O-Cu- to improve the stability of copper ion release. Furthermore, the Cu-MOF-74/SiO₂ aerogel hybrid material prepared by mixing copper ions with SiO₂ aerogel first makes copper ions better fixed on aerogel, which makes the stable release performance of copper ions in these coating (CuSi coatings) much better than that of the simple mixed coating (Cu + Si coatings). Moreover, with the increase of the content of Cu-MOF-74/SiO₂ aerogel hybrid material, the release rate of copper ions is increasingly stable (fluctuating between 0.07–0.10 mg/L), because the increase of -Si-O- bonds from hybrid material makes it easier for hybrid material to combine with PDMS through -Si-O- bond, and improves the bonding between -Si-O-Cu- in hybrid material and -Si-O- in polymer material (Fig. 10). In general, the prepared coating containing the hybrid material can stably and continuously release Cu ions. Finally, it is noted that the Blank coating also has a small amount of copper ion leaching in the early stage, which may be due to impurities in the raw materials used to prepare PDMS.

Generally, the hydrolytic stability of MOF-74 is good, but water is able to penetrate their channel and destroy the structure. As shown in Eq. (1), water molecules enter the channel at the beginning, then some water molecules interact with exposed Cu ions and dissociate into OH⁻ and H⁺. OH⁻ and H⁺ bond to the Cu²⁺ and O atoms of DHBDC⁴⁻, respectively. Finally, the MOFs framework collapse with the dissociation of Cu-O bonds, then releasing metal ions [13,56–58]. Cu ions can inactivate the main enzymes of marine organisms relying on for

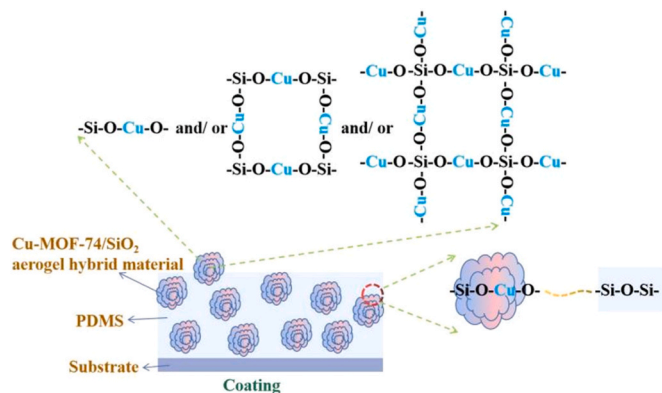


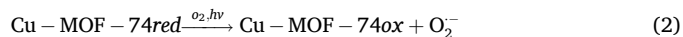
Fig. 10. Schematic diagram of Cu-MOF immobilized on SiO₂ aerogel in Cu-MOF/SiO₂ aerogel@PDMS coating.

survival, or cause the protein of biological cells to flocculate and produce metalloprotein precipitation, leading to changes in biological tissues, so as to achieve antifouling [59–61]. Finally, it is noted that the concentration of copper ions released is relatively low (<0.21 mg/L), which is in line with our expectations, as the cumulative effect of copper ions may have a negative impact on marine ecology. Therefore, coatings releasing less or even no copper ions will contribute to environmental protection.



3.4.2. The production of $\cdot\text{OH}$

The oxidative stress induced by reactive oxygen species (ROS) imbalance is considered one of the main antibacterial mechanisms for Cu-MOF-74 [29,62,63]. To assess the ROS generation by Cu-MOF-74/ SiO_2 aerogel hybrid material, $\cdot\text{OH}$ was detected because $\cdot\text{OH}$ is an important active product of ROS [62]. As shown in Fig. 11a, the Blank coating cannot be detected the signal of $\cdot\text{OH}$. Coatings containing Cu-MOF all have the signals, and the signals of the Control, Cu + Si 6 %, and CuSi 6 % coatings increase sequentially due to the increase of Cu-MOF content. This indicates that Cu-MOF-74/ SiO_2 aerogel hybrid material can promote an increase in reactive oxygen species content. In addition, as the reaction time prolongs, the content of free radicals generated increases (Fig. 11b–d). Literature reported that Cu-MOF-74 can react with oxygen to form superoxide ions (O_2^-), then the generated O_2^- will react with H^+ in the solution to produce H_2O_2 , and finally the H_2O_2 reacts with H^+ to form $\cdot\text{OH}$ (Eqs. (2)–(4)) [27,62,63]. ROS work on antimicrobial properties by interacting with key cellular components, such as proteins, lipids and nucleic acids to produce toxic effects and then inactivate the bacteria [64–67].



3.4.3. Morphology of *E. coli*, calcareous tubeworms and coatings

The antibacterial mechanism of Cu-MOF reported in literature includes the release of Cu ions, cell wall and plasma membrane rupture, the presence of Cu open metal sites and oxidative stress by ROS unbalance [12,62,68,69]. As can be seen from Fig. 12, after interacting with the coating without Cu-MOF (Blank: only PDMS), the morphology of *E. coli* remains intact. However, after interacting with the coatings containing Cu-MOF (Control, Cu + Si 6 % and CuSi 6 % coating), most bacterial cells are damaged, with cells appearing as concave or flattened. The destruction of important proteins and enzymes by copper ions mentioned above, as well as the oxidative damage of proteins, lipids, and nucleic acids by ROS, and the rupture of the cell wall and plasma membrane by contacting with the Cu^{2+} open metal sites (act as the oxidizer for the organic moieties), ultimately manifest as physical damage to bacterial cells, leading to a large outflow of cell contents and bacterial death (Fig. 14) [8,12,63].

Macrofouling organisms (e.g., kelp, bryozoans, mussels, barnacles, calcareous tubeworms) bring the majority of weight and/or hydrodynamic to marine facilities such as ships [70,71]. It can be seen from Figs. 8 and 13a, only calcareous tubeworms grow on the PDMS coatings (with or without Cu-MOF-74), indicating that these as-prepared coatings can prevent the colonization of most macro-fouling organisms (Fig. 14). Fig. 13b shows that the body of the calcareous tubeworm has also been damaged, suggesting that Cu-MOF-74 can not only destroy bacterial cells, but also cause damage to macrofouling organisms. In addition, we

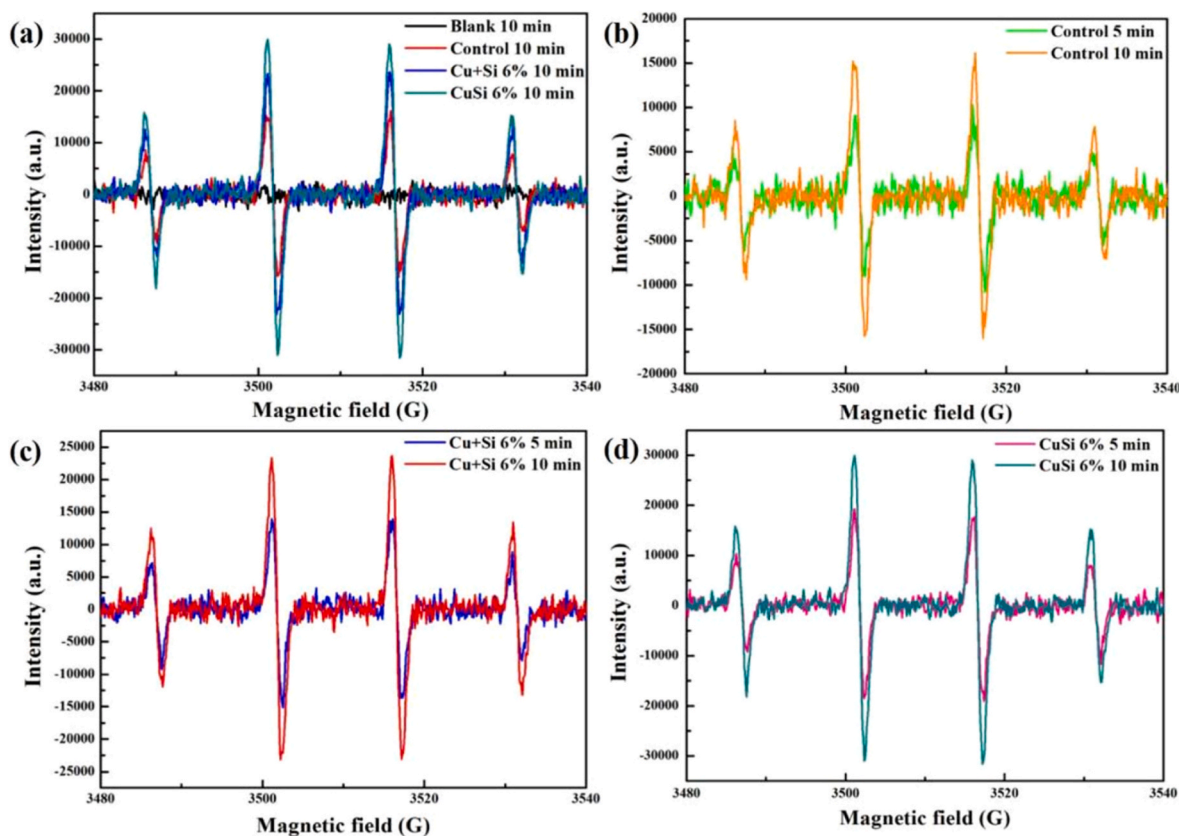


Fig. 11. (a) The ROS (represented by $\cdot\text{OH}$) signals of the Blank, Control, Cu + Si 6 % and CuSi 6 % coatings after detected for 10 min. The intensity of $\cdot\text{OH}$ signals obtained by the (b) Control coating, (c) Cu + Si 6 % coating and (d) CuSi 6 % coating at different detection times (5 min and 10 min).

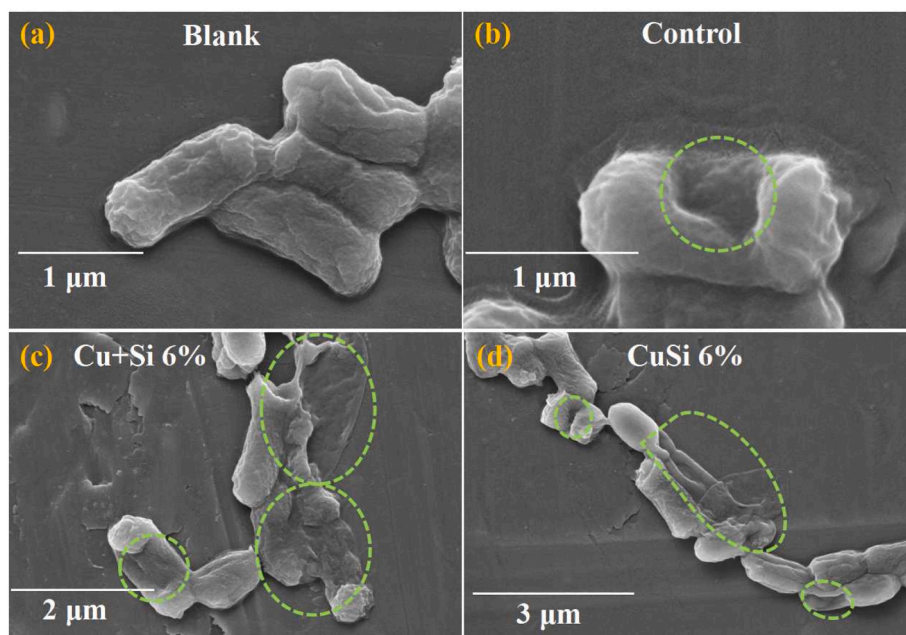


Fig. 12. Morphology of *E. coli* after contacting with (a) Blank, (b) Control, (c) Cu + Si 6 % and (d) CuSi 6 % coating for 24 h.

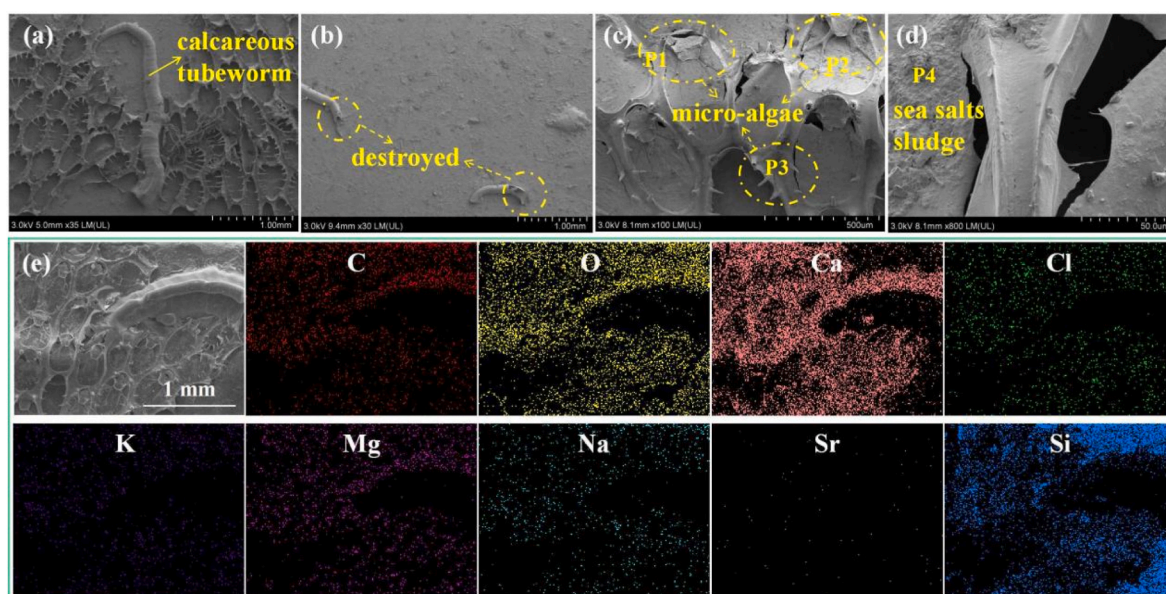


Fig. 13. SEM and Mapping images of calcareous tubeworms and the network-like substance on CuSi 3 % coating.

also observe the network-like substances on the coating from Fig. 13a–c–e. After zooming in, it is found that the substances at positions 1 (P1) and 2 (P2) are closer to some calcified micro-algae, which had long trailing stalks as described in the literature [72]. The presence of thorns at positions 3 (P3) and other locations, as well as the abundance of Ca element (Fig. 13c–e and Table S1) in the network-like substances, further confirms that these substances may be some thorny algae [73, 74]. Substances such as at position 4 (P4) and elements such as Cl, K, Mg, Na and Sr suggest that these network-like substances may also contain some marine salts and sludge. However, it can be seen Fig. 8b that the distribution of these substances on the coatings containing Cu-MOF is less than that of the blank group (only PDMS), indicating that Cu-MOF also has some resistance to them, but further research is needed for complete defense. Besides, these micro-algae and sludge may facilitate further colonization of fouling organisms, supporting our speculation in

section 3.3.1.

4. Conclusions

We successfully prepared the Cu-MOF/SiO₂ aerogel hybrid material which was filled in organic silicon coating (Cu-MOF/SiO₂ aerogel@PDMS coating) to enhance the static marine anti-biofouling performance. The coating has continuous antibacterial performance as it could continuously provide copper ions, open Cu sites and ROS. The enhancement of hydrophobicity And the presence of SiO₂ aerogel is helpful to immobilize Cu ions (Cu-MOF) by the formation of -Si-O-Cu-O-, thereby alleviating the massive release of copper ions to seawater. In addition, the Cu-MOF/SiO₂ aerogel@PDMS coating has good elastic recovery ability, which is conducive to preventing the coating from being easily damaged by the impact of sediment in the ocean. Moreover,

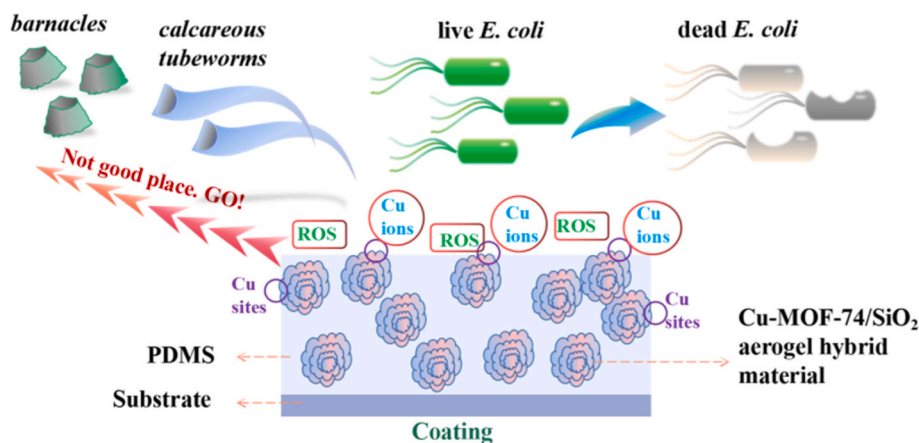


Fig. 14. Schematic diagram of the anti-biofouling mechanisms of the coating.

it can enhanced the long-term static anti-biofouling performance of PDMS in the local sea by effectively preventing the colonization of macrofouling organisms (e.g. barnacles, calcareous tubeworms).

CRediT authorship contribution statement

Liuqin Zhang: Writing – review & editing, Writing – original draft, Supervision, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Huali Li:** Supervision, Funding acquisition, Formal analysis. **Gege Zhang:** Software, Investigation. **Qiuping Li:** Visualization, Software, Investigation, Data curation. **Fa-Qian Liu:** Supervision, Project administration, Methodology, Funding acquisition.

Notes

The authors declare no competing financial interest.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.mtchem.2025.102918>.

Data availability

Data will be made available on request.

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